Class 2, 3.

W-4 (Probabilistic Reasoning)

Lecture Overview

- Probability Theory Review: Basics of probability, Bayes' theorem.
- Bayesian Networks: Representation and inference.
- Hidden Markov Models (HMMs): Applications in time-series data.

Overview of Probability Theory.

- Objective: Understand randomness in real-world models.
- Key Point: Probabilistic models deal with uncertainties in phenomena.

Sample Space and Events.

- Definition: The sample space S represents all possible outcomes of an experim
- Example:

(1) If the experiment consists of the flipping of a coin, then

$$S = \{H, T\}$$

where H means the outcome of the toss is a head and T is a tail.

(2) If the experiment consists of rolling a die, then the sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

where the outcome of i means that i appeared on the die; $i = 1, 2, 3, 4, 5,$

③ If the experiment consists of flipping two coins, then the sample space consists of the following four points:

$$S = \{(H,H), (H,T), (T,H), (T,T)\}$$

The outcome will be (H,H) if both coins come up heads; it will be (H,T) if the first coin comes up heads and the second coin comes up tails; it will be (T,H) if the first comes up tails and the second heads; and it will be (T,T) if both coins come up tails.

④ If the experiment consists of rolling two dice, then the sample space consists of the following 36 points:

$$S = \left\{ \begin{array}{llllll} (1,1), & (1,2), & (1,3), & (1,4), & (1,5), & (1,6), \\ (2,1), & (2,2), & (2,3), & (2,4), & (2,5), & (2,6), \\ (3,1), & (3,2), & (3,3), & (3,4), & (3,5), & (3,6), \\ (4,1), & (4,2), & (4,3), & (4,4), & (4,5), & (4,6), \\ (5,1), & (5,2), & (5,3), & (5,4), & (5,5), & (5,6), \\ (6,1), & (6,2), & (6,3), & (6,4), & (6,5), & (6,6) \end{array} \right\}$$

where the outcome (i,j) is said to occur if i appears on the first die and j on the second die.

⑤ If the experiment consists of measuring the lifetime of a car, then the sample space consists of all nonnegative real numbers. That is,

$$S = [0,\infty)^*$$

CamScanner

- **Definition:** Any subset E of the sample space S is known as an event.

- Example:

(1') In Example (1), if $E = \{H\}$, then E is the event that a head appears on the flip of the coin. Similarly, if $E = \{T\}$, then E would be the event that a tail appears.

(2') In Example (2), if $E = \{1\}$, then E is the event that 1 appears on the roll of the die. If $E = \{2,4,6\}$, then E would be the event that an even number appears on the roll.

(3') In Example (3), if $E = \{(H,H),(H,T)\}$, then E is the event that a head appears on the first coin.

(4') In Example (4), if $E = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$, then E is the event that the sum of the dice equals seven.

(5') In Example (5), if $E = \{(2,6)\}$, then E is the event that the car lasts between two and six years.

CamScanner

Union and Intersection of Events.

- For any 2 events E and F of a sample space S we define the new event E∪F to consist of all outcomes that are either in E or in F or in both E and F. That is, the event E∪F will occur if either E or F occurs. For example, if E = {H} and F = {T}, then

$$E \cup F = \{H, T\}.$$

That is, E∪F would be the whole sample space S. If E = {1,3,5}, and F = {1,2,3}, then

$$E \cup F = \{1,2,3,5\}.$$

and thus E∪F would occur if the outcome of the die is 1 or 2 or 3 or 5. The event E∪F is often referred to as the union of the event E and F.

- For any 2 events E and F, new event is EF or E∩F. The event EF will occur only if both E and F occur. If E = {1,3,5}, and F = {1,2,3}, then

$$EF = \{1,3\}.$$

and thus EF would occur if the outcome of the die is either 1 or 3.

→ Null event is denoted by ϕ.

if E = {T}, and F = {H}, EF = ϕ = event consisting of no outcomes.

If EF = ϕ, then E and F are said to be mutually exclusive.

CamScanner

Probabilities Defined on Events.

- Consider an experiment whose sample space is S. For each event E of the sample space S, we assume that a number $P(E)$ is defined and satisfies the following 3 conditions:

(i) $0 \leq P(E) \leq 1$.

(ii) $P(S) = 1$.

(iii) For any sequence of events $E_1, E_2, \ldots$ that are mutually exclusive, that is, events for which $E_n E_m = \phi$ when $n \neq m$, then,

$$P\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P(E_n)$$

We refer to $P(E)$ as the probability of the event E.

- **Example 1**

In the coin tossing example, if we assume that a head is equally likely to appear as a tail, then we would have,

$$P(\{H\}) = P(\{T\}) = \frac{1}{2}.$$

Example 2.

In the die tossing example, if we supposed that all six numbers were equally likely to appear, then we would have

$$P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = P(\{5\}) = P(\{6\}) = \frac{1}{6}.$$

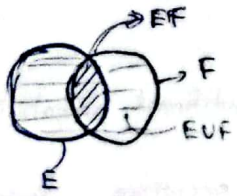
Moreover, the probability of getting an even number would equal

$$P(\{2,4,6\}) = P(\{2\}) + P(\{4\}) + P(\{6\})$$
$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}.$$

CamScanner

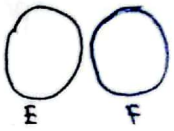
- Derive a formula for $P(E \cup F)$, the probability of all outcomes either in E or in F

$$P(E) + P(F) = P(E \cup F) + P(EF)$$



- Non-overlapping events (Mutually exclusive): If E and F are mutually exclusive (i.e., they cannot happen at the same time, so $P(EF) = 0$), then:

$$P(E) + P(F) = P(E \cup F)$$



- Example 1.3

Suppose we toss 2 coins, and suppose that we assume that each of the four outcomes in the sample space.

$$S = \{(H,H), (H,T), (T,H), (T,T)\}.$$

is equally likely and hence has probability $\frac{1}{4}$. Let,

$$E = \{(H,H), (H,T)\} \text{ and } F = \{(H,H), (T,H)\}.$$

$P(E \cup F)$, the probability that either the first or the second coin falls heads, is given by,

$$P(E \cup F) = P(E) + P(F) - P(EF)$$

$$= \frac{2}{4} + \frac{2}{4} - \frac{1}{4}$$

$$= \frac{3}{4}$$

This probability could, of course, have been computed directly since

$$P(E \cup F) = P(\{(H,H), (H,T), (T,H)\}) = \frac{3}{4}.$$

A.

Conditional Probabilities

- Conditional probability, denoted as $P(A|B)$, represents the probability of an event A occuring, given that another event B has already occurred.

The conditional probability of A given B, where $P(B) > 0$, is mathematically defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

where:

- $P(A|B)$ is the conditional probability of A happening given that B has occurred.

Example:

- We roll 2 dice, and we want to find the conditional probability that the sum of the 2 dice is 9, given that the first die shows a 4.

Step1: Problem definition.

We are dealing with 2 events:

Event A: The sum of the 2 dice equals 9.

Event B: The first die shows 4.

We need to find $P(A|B)$, the probability that the sum of the 2 dice is 9 given that the first die shows 4.

Step2: Total Possible Outcomes

Since each die has 6 faces, the total number of possible outcomes: $6 \times 6 = 3$

So, the sample space S consists of 36 possible outcomes:

$$S = \{(1,1), (1,2), \ldots, (6,6)\}.$$

Step 3: Event A: The sum is 9

$A = \{(3,6), (4,5), (5,4), (6,3)\}$

$P(A) = \frac{4}{36} = \frac{1}{9}$

Step 4: Event B: the first Die is 4

$P(B) = \frac{6}{36} = \frac{1}{6}$.

Step 5: Event $A \cap B$: The sum is 9 and the first Die is 4

$A \cap B = \{(4,5)\}$.

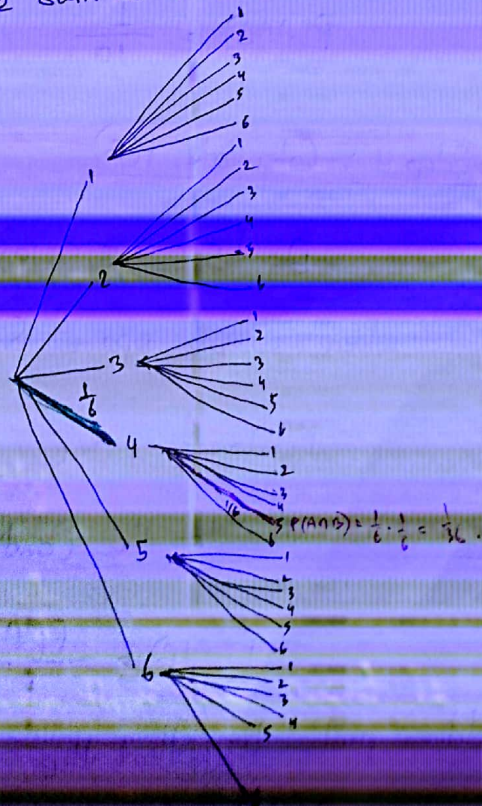
$P(A \cap B) = \frac{1}{36}$

Step 6: Calculate the conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{36}}{\frac{1}{6}} = \frac{1}{36} \cdot 6 = \frac{1}{6}.$$

Thus, the conditional ~~probabilities~~ probability that the sum of the two dice is 9, given that the first die is 4, is:

$$P(A|B) = \frac{1}{6}.$$



CamScanner

and "then one of the cards is drawn. If we are told that the number on the drawn card is at least five, what is the conditional probability that it is ten?

Step 1: Define the events.

Step 2: Calculate the Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

The conditional probability that the card is 10, given that the card is at least 5, is: $P(A|B) = \frac{1}{6}$.

Bayesian Classification

(ML Method for Supervised Learning)

- **A statistical classifier:** performs probabilistic prediction, i.e., predicts class membership probabilities.
- **Foundation:** Based on Bayes' Theorem.
- **Performance:** A simple Bayesian classifier, naïve Bayesian classifier, has comparable performance with decision tree and selected neural network classifiers.
- **Incremental:** Each training example can incrementally increase/decrease the probability that a hypothesis is correct - prior knowledge can be combined with observed data.
- **Standard:** Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which others methods can be measured.

Bayes' Theorem: Basics.

- Total probability Theorem: $P(B) = \sum_{i=1}^{M} P(B|A_i) \cdot P(A_i)$
- Bayes' Theorem: $P(H|X) = \frac{P(X|H) \cdot P(H)}{P(X)} = \frac{P(X|H) \cdot P(H)}{P(X)}$

  - Let X be a data sample ("evidence"): class label is unknown
  - Let H be a hypothesis that X belongs to class C.
  - Classification is to determine $P(H|X)$, (i.e., posteriori probability): the probability that the hypothesis holds given the observed data sample X.
  - $P(H)$ (prior probability): the initial probability.
  - E.g., X will buy computer, regardless of age, income, ...
  - $P(X)$: probability that sample data is observed.
  - $P(X|H)$ (likelihood): the probability of observing the sample X, given that the hypothesis holds. E.g., Given that X will buy computer, the probability that X is 31...40,

medium income.

Bayes' Theorem Derivation

Derivation of Bayes' Theorem.

Bayes' Theorem is derived from the definition of conditional probability. By definition,

$$P(H|E) = \frac{P(H \cap E)}{P(E)}$$

This represents the probability of H and E occuring together, normalized by the probability of the evidence E. Similarly, the joint probability $P(H \cap E)$ can also be expressed as:

$$P(H \cap E) = P(E|H) \cdot P(H)$$

$$\therefore P(E|H) = \frac{P(H \cap E)}{P(H)}$$

This gives us the joint probability of both H and E occuring. Substituting this into the equation for conditional probability, we get:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Law of Marginal Likelihood:

The denominator in Bayes' Theorem, $P(E)$, is called the ~~model~~ marginal likelihood or the evidence. It represents the overall probability of observing the evidence E across all possible hypotheses. If we have several mutually exclusive and exhaustive hypotheses $H_1, H_2, \ldots, H_n$, the marginal likelihood is calculated using the law of total probability:

$$P(E) = \sum_{i=1}^{n} P(E|H_i) \cdot P(H_i)$$

In other words, P(E) is the weighted sum of the likelihoods of E under all possible hypotheses, where each likelihood is weighted by the ~~priori~~ prior probability of the corresponding hypothesis.

Prediction Based on Bayes' Theorem

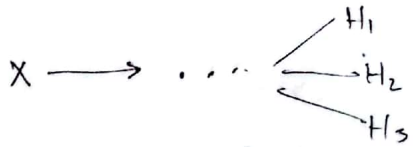
- Given training data X, posteriori probability of a hypothesis H, P(H|X), follows the Bayes' theorem.

$$P(H|X) = \frac{P(X|H)\,P(H)}{P(X)}$$

- Informally, this can be viewed as

posteriori = likelihood x prior / evidence.

- Predicts X belongs to C_i iff the probability $P(C_i|X)$ is the highest among all the $P(C_k|X)$ for all the k classes.
- Practical difficulty: It requires initial knowledge of many probabilities, involving significant computational cost



Classification is to derive the maximum posteriori

- Let D be a training set of tuples and their associated class labels, and each tuple is represented by an n-D attribute vector $X = (x_1, x_2, \ldots, x_n)$
- Suppose there are m classes, $C_1, C_2, \ldots, C_m$
- Classification is to derive the maximum posteriori, i.e., the maximal $P(C_i|X)$
- This can be derived from Bayes' Theorem:

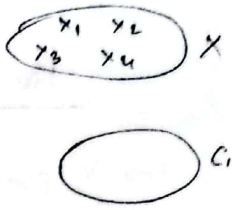
$$P(C_i|X) = \frac{P(X|C_i) \cdot P(C_i)}{P(X)}$$

- Since P(X) is constant for all classes, only $\boxed{P(C_i|X) = P(X|C_i) \cdot P(C_i)}$ needs to be maximized.

Naive Bayes Classifier.

- A simplified assumption: attributes are conditionally independent (i.e., no dependence relation between attributes):

$$P(X|C_i) = \prod_{k=1}^{n} P(x_k|C_i) = P(x_1|C_i) \times P(x_2|C_i) \times \dots \times P(x_n|C_i)$$



$$P(C_i|X) = P(X|C_i) \cdot P(C_i)$$

- This greatly reduces the computation cost: Only counts the class distribution

Naive Bayes Classifier: Training Dataset

age	income	student?	credit_rating	buys_computer?
<=30	high=h	no=n	fair=f	no=n
<=30	h	n	excellent=ex	n
31...40	h	n	f	yes=y
>40	medium=m	n	f	y
>40	low=l	yes=y	f	y
>40	l	y	ex	n
31...40	l	y	ex	y
<=30	m	n	f	n
<=30	l	y	f	y
>40	m	y	f	y
<=30	m	y	ex	y
31...40	m	n	ex	y
31...40	h	y	f	y
>40	m	n	ex	n

Class:

C1: buys_computer = 'yes'

C2: buys_computer = 'no'

Data to be classified:

X = (age <= 30,

Income = medium,

Student = yes,

Credit_rating = Fair)

- P(Ci):

$P(\text{buys_computer} = \text{"yes"}) = \frac{9}{14} = 0.643$

$P(\text{buys_computer} = \text{"no"}) = \frac{5}{14} = 0.357$

- Compute P(X|Ci) for each class

$P(\text{age} <= 30 \mid \text{buys_computer} = \text{"yes"}) = \frac{2}{9} = 0.222$ (PLAN B)

$P(\text{age} <= 30 \mid \text{buys_computer} = \text{"no"}) = \frac{3}{5} = 0.6$

$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{'yes'}) = \frac{4}{9} = 0.444$

$P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{'no'}) = \frac{2}{5} = 0.4$

$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{'yes'}) = \frac{6}{9} = 0.667$

$P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{'no'}) = \frac{1}{5} = 0.2$

$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"yes"}) = \frac{6}{9} = 0.667$

$P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"no"}) = \frac{2}{5} = 0.4$

We know,

$P(X|C_i) = P(x_1|C_i) \cdot P(x_2|C_i) \cdots P(x_n|C_i)$. So, when, $C_i \Rightarrow$ buys_computer = "yes" and X is data,

$P(X|C_i) = P(\text{age} \le 30 \mid \text{buys_computer} = \text{"yes"}) \cdot P(\text{income} = \text{"medium"} \mid \text{buys_computer} = \text{"yes"}) \cdot P(\text{student} = \text{"yes"} \mid \text{buys_computer} = \text{"yes"}) \cdot P(\text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"yes"})$

Therefore,

$P(\text{age} <= 30, \text{income} = \text{"medium"}, \text{student} = \text{"yes"}, \text{credit_rating} = \text{"fair"} \mid \text{buys_computer} = \text{"yes"})$

$= (0.222) \cdot (0.444) (0.6667) \cdot (0.6667) = \boxed{0.0439}$

Now, $P(X \mid \text{buys_computer} = \text{"no"}) = (0.6)(0.4)(0.2)(0.4) = \boxed{0.0192}$

We know, for classification,

$P(C_i|X) \propto P(X|C_i)P(C_i)$ [$\because P(X)$ is constant]

maximum $P(C_i|X)$ determines X's belonging class. Hence,

$P(X \mid \text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = (0.0439) \cdot (0.643) = 0.0282$

$P(X \mid \text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = (0.0192) \cdot (0.357) = 0.0068544$

So, $P(\text{buys_computer} = \text{"yes"} \mid X) > P(\text{buys_computer} = \text{"no"} \mid X)$

Therefore, X belongs to class ("buys_computer = "yes").

(W-4 - Probabilistic Reasoning - (1-37) and (1-57).

From Bayesian Inference for AI

(8-

Naive Bayes Classification with Laplacian Correction.

Let's consider a small example of a Naive Bayesian Classifier where we predict whether a person will buy a computer based on their income and student status. The dataset is small, and in this case, we need to use Laplacian correction (also known as Laplace smoothing) to handle cases where we have zero counts in the conditional probabilities.

1 Dataset

We consider the following dataset where we predict whether a person will buy a computer based on their income and student status.:

Table 1: Dataset for Buy Computer Class Prediction

ID	Income	Student	Buys Computer (Class)
1	High	No	No
2	High	No	No
3	Medium	No	Yes
4	Low	Yes	Yes
5	Low	Yes	Yes
6	Low	No	No.

We aim to predict whether a person with ~~high~~ Income High Income and Students Status = Yes ~~not~~ will buy a computer .

CamScanner

2 Steps to Perform Naive Bayes Classification.

P(C|X)
• P(X|C) · P(C)

P(X|C) · P(X|C)

2.1 Prior Probabilities

The prior probabilities for each class are calculated as follows:

$$P(\text{Yes}) = \frac{\text{Number of people who bought a computer}}{\text{Total number of people}} = \frac{3}{6} = 0.5$$

$$P(\text{No}) = \frac{3}{6} = 0.5$$

2.2 Likelihoods Without Laplacian Correction

Next, we ~~feature~~ calculate the likelihoods for each feature given the class.

2.2.1 Income

For the class Yes (Buys Computer):

$$P(\text{Income} = \text{High} \mid \text{Yes}) = \frac{P(\text{High and Yes})}{P(\text{Yes})} = \frac{\frac{0}{6}}{\frac{3}{6}} = 0$$

For the class No (Does not buy computer):

$$P(\text{Income} = \text{High} \mid \text{No}) = \frac{2}{3} = 0.667.$$

2.2.2 Student Status

For the class Yes:

$$P(\text{Student}) \cdot P(\text{Student} = \text{Yes} \mid \text{Yes}) = \frac{2}{3}$$

(2 → # of student & Yes; 3 → # of Yes)

For the class No:

$$P(\text{Student} = \text{Yes} \mid \text{No}) = \frac{0}{3} = 0.$$

2.3 Laplacian Correction.

To avoid zero probabilities, we apply Laplacian correction by adding 1 to the counts.

2.3.1 Corrected likelihoods for income (with k = 3)

↳ since income → high, medium, low.

For the class Yes:

$$P(\text{Income} = \text{High} \mid \text{Yes}) = \frac{0+1}{3+3} = \frac{1}{6}$$

(Yes; $k = 3$)

$P(X_1 = \text{High} \mid C_1) = \frac{1}{6}$ (Yes; ↳ income)

For the class No:

$$P(\text{Income} = \text{High} \mid \text{No}) = \frac{2+1}{3+3} = \frac{3}{6} = \frac{1}{2}.$$

2.3.2 Corrected likelihoods for student status (with k = 2)

(2; student? → Yes, No)

For the class Yes:

$$P(\text{Student} = \text{Yes} \mid \text{Yes}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$$

For the class No:

$$P(\text{Student} = \text{Yes} \mid \text{No}) = \frac{0+1}{3+2} = \frac{1}{5} = 0.2.$$

2.4 Posterior Probabilities.

Using the corrected likelihoods, we calculate the posterior probabilities for each class using the Naïve Bayes formula:

2.4.1 For Class = Yes.

$P(C_i \mid X) = P(C_i) \cdot P(X \mid C_i)$

$$P(\text{Yes} \mid \text{High Income}, \text{Student} = \text{Yes}) = P(\text{Yes}) \cdot P(\text{Income} = \text{High} \mid \text{Yes}) \cdot P(\text{Student} = \text{Yes} \mid \text{Yes})$$

$$= 0.5 \times 0.167 \times 0.6 = 0.0501$$

2.4.2 For class = No.

$P(\text{No} \mid \text{High Income, Student= Yes}) = P(\text{No}) \cdot P(\text{High Income} \mid \text{No}) \cdot P(\text{Student= Yes} \mid \text{No})$

$= 0.5 \times 0.5 \times 0.2$

$= 0.05.$

2.5 Final Prediction.

Since $P(\text{Yes} \mid \text{High Income, Student = Yes}) = 0.0501$ is greater than $P(\text{No} \mid \text{High Income, Student= Yes}) = 0.05$, the model predicts that the person will buy a computer.

CamScanner

* Effect of using k in Laplacian Correction for Naive Bayesian Classification.

In Naive Bayesian classification, Laplacian correction (also known as Laplace smoothing) is used to handle zero probabilities when a feature value does not appear in the training data for a particular class. The correction prevents the entire probability from # becoming zero when multiplying the probabilities of feature values. A key aspect of Laplacian correction is the use of k, the number of possible values a feature can take. Following demonstrates the effect of not using k in Laplacian correction and how it can result in incorrect probabilities.

(1) Problem Statement

Consider a feature called Weather with 3 possible values:

- Sunny
- Rainy
- Cloudy.

We are predicting whether people will go Outdoors (Yes/No) based on the weather. The training dataset is shown # below:

Table 1: Training Dataset

ID	Weather	Outdoors (Class)
1	Sunny	Yes
2	Sunny	Yes
3	Rainy	Yes
4	Sunny	No
5	Rainy	No
6	Sunny	No.

CamScanner

1.1 Without Laplacian Correction.

The probabilities for the weather conditions given "Outdoors = Yes" before any smoothing are:

$$P(\text{Sunny}|\text{Yes}) = \frac{2}{3}, \quad P(\text{Rainy}|\text{Yes}) = \frac{1}{3}, \quad P(\text{Cloudy}|\text{Yes}) = 0$$

(2 = # of sunny and yes; 3 = # of yes.)

The sum of the probabilities is:

$$P(\text{Sunny}|\text{Yes}) + P(\text{Rainy}|\text{Yes}) + P(\text{Cloudy}|\text{Yes}) = \frac{2}{3} + \frac{1}{3} + 0 = 1.$$

However, the zero probability for Cloudy will cause problems if this value is encountered during classification.

1.2 With Laplacian Correction (Without k)

Using Laplacian correction without k (i.e. adding 1 to the numerator and denominator), we calculate the probabilities as:

$$P(\text{Sunny}|\text{Yes}) = \frac{2+1}{3+1} = \frac{3}{4}, \quad P(\text{Rainy}|\text{Yes}) = \frac{2}{4}, \quad P(\text{Cloudy}|\text{Yes}) = \frac{1}{4}$$

The sum of the probabilities becomes:

$$P(\text{Sunny}|\text{Yes}) + P(\text{Rainy}|\text{Yes}) + P(\text{Cloudy}|\text{Yes}) = \frac{6}{4} = 1.5.$$

This is greater than 1, which violates the rules of probability.

1.3 With Laplacian Correction (With k=3)

Now, using Laplacian correction with k=3 (since there are three possible values for Weather: Sunny, Rainy, Cloudy), we recalculate the probabilities:

$$P(\text{Sunny}|\text{Yes}) = \frac{2+1}{3+3} = \frac{3}{6} = 0.5, \quad P(\text{Rainy}|\text{Yes}) = \frac{1+1}{3+3} = \frac{2}{6} = \frac{1}{3}, \quad P(\text{Cloudy}|\text{Yes}) = \frac{0+1}{3+3} = \frac{1}{6}.$$

CamScanner

The sum of the probabilities becomes :

$$P(\text{Sunny} \mid \text{Yes}) + P(\text{Rainy} \mid \text{Yes}) + P(\text{Cloudy} \mid \text{Yes}) = 0.5 + \frac{1}{3} + \frac{1}{6} = \frac{6}{6} = 1 .$$

(2) Conclusion

The use of k in Laplacian correction is essential for maintaining the sum of probabilities as 1. Without k, the smoothing effect is not properly distributed across all possible values of the feature, leading to overestimated probabilities. By including k, we ensure that the probabilities are correctly normalized and that unseen values do not disproportionately influence the classification.

(?) Naive Bayes Classification with Laplacian Correction (Practice Example)

(1) Problem Description

You are tasked with building a Naive Bayesian classifier to predict whether a student will pass or fail a course based on the following features :

- Attendance (low, medium, high)
- Study Hours (low, medium, high)
- Previous Grades (pass, fail)

The dataset is as follows :

(Data set is in next page).

(2) Questions.

(1) Calculate the prior probabilities for both classes (pass/fail)

(2) Calculate the conditional probabilities for each feature given the class, using laplacian correction to handle any zero probabilities.

(3) Classify the following student using the Naive Bayesian algorithm:

- Attendance : Medium
- Study hours : Medium
- Previous Grades : Pass.

(4) Compare the result with the original dataset and discuss how laplacian correction helps prevent issues with zero probabilities.

(3) Expected Learning Outcomes.

Students will learn how to handle small datasets with zero probabilities by applying laplacian correction.

CamScanner

Table 1: Dataset for Pass/Fail Prediction

ID	Attendance	Study Hours	Previous Grades	Pass/Fail (Class)
1	Low	Low	Fail	Fail
2	High	High	Pass	Pass
3	Medium	Low	Pass	Fail
4	Medium	Medium	Fail	Fail
5	High	Medium	Pass	Pass
6	Low	High	Fail	Fail
7	Medium	Medium	Pass	Pass
8	Low	Low	Fail	Fail
9	High	High	Pass	Pass
10	Medium	Low	Fail	Fail

Answer Q1:

- Pass: 4 students
- Fail: 6 students.

Total = 10 students.

$$P(\text{Pass}) = \frac{4}{10} = 0.4 \quad , \quad P(\text{Fail}) = \frac{6}{10} = 0.6.$$

Answer to Q2:

- For Attendance, k = 3. *For Pass Yes class,

$$P(\text{Attendance} = \text{Low} \mid \text{Yes}) = \frac{0+1}{4+3} = \frac{1}{7} = 0.143$$

$$P(\text{Attendance} = \text{Medium} \mid \text{Pass}) = \frac{1+1}{7} = 2/7 = 0.286$$

$$P(\text{Attendance} = \text{High} \mid \text{Pass}) = \frac{3+1}{7} = \frac{4}{7} = 0.571$$

* For Fail class:

$$P(\text{Attendance} = \text{Low} \mid \text{Fail}) = \frac{3+1}{6+3} = \frac{4}{9} = 0.444$$

$$P(\text{Attendance} = \text{Medium} \mid \text{Fail}) = \frac{3+1}{6+3} = \frac{4}{9} = 0.444.$$

CamScanner

$$P(\text{Attendance} = \text{Medium/High} \mid \text{Fail}) = \frac{0+1}{9} = \frac{1}{9} = 0.111$$

- For study hours, $k = 3$ (Low, High, Medium)

  * For class Pass,

$$P(\text{Study Hours} = \text{Low} \mid \text{Pass}) = \frac{0+1}{4+3} = \frac{1}{7} = 0.143$$

$$P(X_2 = \text{Medium} \mid \text{Pass}) = \frac{2+1}{4+3} = \frac{3}{7} = 0.429$$

$$P(X_2 = \text{High} \mid \text{Pass}) = \frac{2+1}{4+3} = \frac{3}{7} = 0.429$$

  * for class Fail,

$$P(\text{Previous Grades} = X_2 = \text{Low} \mid \text{Fail}) = \frac{4+1}{6+3} = \frac{5}{9} = 0.556$$

$$P(X_2 = \text{Medium} \mid \text{Fail}) = \frac{1+1}{6+3} = \frac{2}{9} = 0.222$$

$$P(X_2 = \text{High} \mid \text{Fail}) = \frac{1+1}{9} = \frac{2}{9} = 0.222.$$

- For Previous Grades $k = 2$ (Pass, Fail).

  Consider, Previous Grades $= X_3$.

  * For Pass class,

$$P(X_3 = \text{Pass} \mid \text{Pass}) = \frac{4+1}{4+2} = \frac{5}{6} = 0.833$$

$$P(X_3 = \text{Fail} \mid \text{Pass}) = \frac{0+1}{6} = \frac{1}{6} = 0.167$$

  * For Fail class,

$$P(X_3 = \text{Pass} \mid \text{Fail}) = \frac{1+1}{6+2} = \frac{2}{8} = 0.25$$

$$P(X_3 = \text{Fail} \mid \text{Fail}) = \frac{5+1}{6+2} = \frac{6}{8} = 0.75.$$

CamScanner

Answer to Q3 :

We know, $P(C_i|X) \propto P(X|C_i) \cdot P(C_i) = P(C_i) \cdot P(X|C_i)$ (class)

New student's features:

- Attendance = Medium $[X_1]$
- Study Hours = Medium $[X_2]$
- Previous Grades = Pass $[X_3]$

We compute the posterior probability for each class:

* For Pass:

$$P(\text{Pass}|X_1, X_2, X_3) = P(\text{Pass}) \cdot P(X_1|\text{Pass}) \cdot P(X_2|\text{Pass}) \cdot P(X_3|\text{Pass})$$

$$= (0.4)(0.286)(0.429)(0.833)$$

$$= 0.041$$

* For Fail:

$$P(\text{Fail}|X_1, X_2, X_3) = P(\text{Fail}) \cdot P(X_1|\text{Fail}) \cdot P(X_2|\text{Fail}) \cdot P(X_3|\text{Fail})$$

$$= (0.6)(0.444)(0.222)(0.25)$$

$$= 0.014785$$

As, $P(\text{Pass}|X_1, X_2, X_3) > P(\text{Fail}|X_1, X_2, X_3)$,

∴ the new student will pass.

CamScanner

Answer to Question 4

Without Laplacian correction:

- Some conditional probabilities (e.g., P(Attendance=Low|Pass)) would be zero.
- This would cause the entire product in posterior probability to be zero, even if the rest of the features support that class.
- Laplacian smoothing avoids this by giving every possible feature a value a non-zero probability.

$$P(\text{Class}|\text{Features}) \propto P(\text{Class}) \times \prod_i P(\text{Feature}_i|\text{Class})$$

* Naive Bayes Classification for Continuous Values with Laplace Correction.

* In a Naïve Bayesian Classification scenario with continuous values like income, you need to treat the continuous attributes differently from discrete ones. The Gaussian Naïve Bayes model is commonly used for handling continuous features, where each continuous feature is assumed to follow a normal (Gaussian) distribution.

(1) Problem Statement

We will predict whether a person will buy a computer based on their:

- Age (categorical)
- Income (continuous)
- Student status (categorical)

(2) Dataset

Table1: Dataset for Buy Computer Class Prediction

ID	Age	Income	Student	Buys Computer (Class)
1	Youth	300	No	No
2	Youth	400	No	No
3	Middle	100	No	Yes
4	Senior	300	Yes	Yes
5	Senior	200	Yes	Yes
6	Youth	300	Yes	No.

We want to predict whether a person with:

- Age = Senior
- Income = 400
- Student status = Yes

buys a computer.

(3) Steps to Perform Naive Bayes Classification.

(3.1) Calculate Prior Probabilities.

We first calculate the prior probabilities for the target class, i.e., the probability of buying a computer or not.

$$P(\text{Yes}) = \frac{\#\text{ of people who bought a computer}}{\text{Total number of people}} = \frac{3}{6} = 0.5$$

$$P(\text{No}) = \frac{3}{6} = 0.5.$$

(3.2) Likelihood for Categorical Attributes (Age and Student Status).

We calculate the likelihoods for the categorical attributes (Age and Student status) based on the class.

3.2.1 Age:

Since we have 3 categories for the Age attribute (Youth, Middle, and Senior), we set $k = 3$.

- For class = Yes:

There are 2 people with Age = Senior. who bought a computer. Applying laplacian correction,

$$P(\text{Age} = \text{Senior} \mid \text{Yes}) = \frac{2+1}{3+3} = \frac{3}{6} = \frac{1}{2}.$$

- For class = No:

There are zero people with Age = Senior who did not buy a computer. Applying Laplacian correction,

$$P(\text{Age} = \text{Senior} \mid \text{No}) = \frac{0+1}{3+3} = \frac{1}{6} = 0.167.$$

CamScanner

3.2.2 Student Status:

- For Student = Yes given Yes:

$$P(\text{Student} = \text{Yes} \mid \text{Yes}) = \frac{2}{3}$$

[As no zero, no Laplacian correction].

- For Student = Yes given No:

$$P(\text{Student} = \text{Yes} \mid \text{No}) = \frac{1}{3}.$$

(3.3) Likelihood for Continuous Attribute (Income)

For continuous attributes like Income, we assume that they follow a Gaussian (normal) distribution. The probability density function for a Gaussian distribut is:

$$P(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where -

- x is the income value for the new instance (in this case 400)
- μ is the mean income for the class.
- σ is the standard deviation of income for the class.

Let's calculate the likelihood of Income = 400 given the class.

(#) For class = Yes: Mean μ_{Yes} of Income for class Yes:

$$\mu_{Yes} = \frac{300+200+100}{3} = 200$$

- Standard deviation (σ_{Yes}) of Income for class Yes:

$$\sigma_{Yes} = \sqrt{\frac{(300-200)^2+(200-200)^2+(100-200)^2}{3}} = 100.81.64.$$

Now, we calculate the likelihood of Income = 400 given Yes:

$$P(\text{Income}=400 \mid \text{Yes}) = \frac{1}{\sqrt{2\pi (81.64)^2}} e^{-\frac{(400-200)^2}{2(81.64)^2}}$$

$$= 0.000243$$

* For class = No:

- Mean (μ_{No}) of income for class No:

$$\mu_{No} = \frac{300+400+300}{3} = \frac{1000}{3} = 333.33$$

- Standard deviation of (σ_{No}) of income for class No:

$$\sigma_{No} = \sqrt{\frac{(300-333.33)^2+(400-333.33)^2+(300-333.33)^2}{3}} = 47.14$$

Now, we calculate the likelihood of Income = 400 given No:

$$P(\text{Income}=400 \mid \text{No}) = \frac{e^{-\frac{(400-333.33)^2}{2\cdot(47.14)^2}}}{\sqrt{2\times\pi\times(47.14)^2}} = 0.00311$$

(3.4) Calculate Posterior Probabilities.

Now we calculate the posterior probabilities for both classes using the Naïve Bayes Formula:

- For e Class = Yes:

P(Yes | Age = Senior, Income = 400, Student = Yes)

= P(Yes) · P(Age = Senior | Yes) · P(Income = 400 | Yes) · P(Student = Yes | Yes)

= (0.5) (0.5) (0.000243) · (0.67) = 0.0000407

- For Class = No:

P(No | Age = Senior, Income = 400, Student = Yes) = P(No) · P(Age = Senior | No) · P(Income = 400 | No) · P (Student = Yes | No)

= (0.5) (0.167) · (0.00311) · (0.333) = 0.0000865

CamScanner

(3.5) Final Prediction.

Comparing the posterior probabilities:

- $P(\text{Yes}) = 0.0000407$
- $P(\text{No}) = 0.0000865$.

Since $P(\text{No}) > P(\text{Yes})$, the model predicts that the person will not buy a computer.

(4) Conclusion.

For continuous attributes like Income, the Gaussian Naive Bayes classifier is used, and for categorical attributes, standard probability calculations are performed. Laplacian correction is applied where needed to avoid zero probabilities.

CamScanner

(?) Naïve Bayes Classification with Continuous Values (Practice Example)

(1) Problem Descp Description.

You are given a dataset that contains the following attributes for employees of a company:

- Age (continuous)
- Salary (continuous)
- Education level (High School, Bachelor's, Master's)
- Buys Car (Yes | No)

The dataset is in the next page.

(2) Questions.

① Calculate the ~~priori~~ Prior probabilities for the class "Buys Car" (Yes/No).

② For the continuous attributes Age and Salary, assume they follow a Gaussian distribution. For each class:

- Calculate the mean and standard deviation for Age and Salary.

③ Classify the following employee using the Gaussian Naive Bayesian Algorithm:

- Age: 37
- Salary: 48000
- Education Level: Bachelor's

④ Show your calculations for the pdf of the Gaussian distribution for continuous values of Age and Salary.

⑤ Compare the final result with the original dataset and discuss how continuous features are handled differently from categorical features in Naive Bayes.

(3) Expected Learning Outcomes

Students will gain experience in handling continuous attributes using the Gaussian Naive Bayes approach and understand how to classify data with continuous features.

CamScanner

Table 1: Dataset for Car Purchase Prediction

ID	Age	Salary	Education Level	Buys Car (Class)
1	25	30000	Bachelor's	No
2	35	50000	Masters'	Yes
3	45	70000	B.	Yes
4	28	32000	High School	No
5	55	60000	M.	Yes
6	33	40000	H. S.	No
7	38	55000	B.	Yes
8	30	35000	B.	No
9	50	65000	M.	Yes
10	40	52000	B.	Yes
	$\Sigma = 379$	$\Sigma = 489000$		

Answer to Q1:

Prior Probabilities

Count of "Yes" : 6 | Total: 10

Count of "No" : 4

$P(Yes) = 0.6$ and $P(No) = \frac{4}{10} = 0.4$

Answer to Q2:

age mean, $\mu_{Age} = 37.9$

salary mean, $\mu_{Salary} = 48900$

- Age

For class = Yes,

$$\mu_{Age, Yes} = \frac{35+45+55+38+50+40}{6} = 43.83$$

$$\sigma_{Age, Yes} = \sqrt{\frac{\Sigma(x-\mu)^2}{n}} = \sqrt{\frac{(35-43.83)^2+\cdots+(40-43.83)^2}{6}} = 6.96$$

For class No:

$$\mu_{Age,No} = \frac{25+28+33+30}{4} = 29.$$

$$\sigma_{Age,No} = \sqrt{\frac{(25-29)^2+(28-29)^2+(33-29)^2+(30-29)^2}{4}} \approx 2.94$$

• Salary.

Class: Yes

$$\mu_{Salary,Yes} = \frac{\Sigma(\text{Salary} \cap \text{Yes})}{\#\text{ of Yes}}$$

$$= 58666.67$$

$$\sigma_{Salary,Yes} = 7110.243$$

Class: No

$$\mu_{Salary,No} = \frac{30000+32000+40000+35000}{4}$$

$$= 34250$$

$$\sigma_{Salary,No} = \sqrt{\frac{\Sigma(x-\mu)^2}{4}}$$

$$= 3766.6298$$

Answer to Q3.

• Age = 37, • Salary = 48000 • Education level: Bachelor's

Here, $P(\text{Education level} = \text{Bachelor's} \mid \text{Yes}) = \frac{3+1}{6+3} = \frac{4}{9}$

$P(\text{Education level} = \text{Bachelor's} \mid \text{No}) = \frac{2+1}{6+3} = \frac{3}{9} = \frac{1}{3}$

$$P(x \mid \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$P(Age=37 \mid Yes) = \frac{1}{\sqrt{2\pi(6.96)^2}} e^{-\frac{(37-43.83)^2}{2(6.96)^2}}$$

$$= 0.035$$

$$P(Age=37 \mid No) = \frac{1}{\sqrt{2\pi(2.92)^2}} \times e^{-\frac{(37-29)^2}{2(2.92)^2}} = 0.0032$$

$$P(Salary=48000 \mid Yes) = \frac{e^{-\frac{(48000-58666.67)^2}{2(7110.243)^2}}}{\sqrt{2\times\pi\times(7110.243)^2}} = 6.0000182$$

$$P(Salary=48000 \mid No) = \frac{1}{\sqrt{2\times\pi\times(3766.63)^2}} \times e^{-\frac{(48000-34250)^2}{2\times(3766.63)^2}} = 0.000000135$$

$P(Yes \mid Age=37, Income=48000, Education\ level=Bachelor's)$

$= P(Yes) \cdot P(Age=37 \mid Yes) \cdot P(Salary=48000 \mid Yes)\ P(Education\ level=Bachelor's \mid Yes)$

$= (0.6)\ (0.035)\ (0.0000182)\ \left(\frac{4}{9}\right)$

$= 0.00000016987$

$= 1.6987 \times 10^{-7}$

$P(No \mid Age=37, Salary=48000, Education\ level=Bachelors)$

$= P(No) \cdot P(Age=37 \mid No)\ P(Salary=48000 \mid No) \cdot P(Education\ level=Bachelors \mid No)$

$= (0.4)\ (0.0032)\ (0.000000135)\ \left(\frac{1}{3}\right)$

$= 5.76 \times 10^{-11}$

As, $1.6987 \times 10^{-7} > 5.76 \times 10^{-11}$, the employee will buy car.

Predicted class => Yes

- Question 4 already answered

Answer to question 5

* Comparing the prediction with the original dataset.

We classified a new employee with Age = 37, Salary = 48000, and Education Level = Bachelor's. Closest matches from the original dataset are, ID 2, 7, and 10. They all bought a car.

Our model also predicted "yes" - indicating strong alignment with the original data.

* In Naive Bayes, the treatment of features depends on whether they are categorical or continuous.

① Categorical Features: Values fall into fixed categories (e.g., "Pass" or "Fail").

→ Calculate probabilities from the frequency counts in the training data.

→ Use Laplacian correction to avoid zero probabilities.

→ Simple, fast, but only works for discrete values.

② Continuous Features: Features like age, income, or salary that take on a range of numeric values.

→ Assume the values follow a Gaussian (Normal) distribution.

→ Estimate the mean (μ) and standard deviation (σ) for each class.

→ Use the Gaussian PDF to compute probabilities.

* Formula (for feature value x):

$$P(x|\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This gives a probability score that reflects how likely a numeric input is under a class distribution.

P.T.O

CamScanner

Feature Type	Categorical	Continuous
Values	Discrete labels (e.g., "Yes", "No")	Numeric range (e.g., 25, 30, 45)
Probability Method	Frequency-based	Gaussian (normal) distribution
Parameters needed	Count tables	Mean (μ), Standard deviation (σ)
Smoothing needed	Laplacian correction	Not needed (unless Gaussian fails)
Example / Use		

Summary:

- Categorical features rely on counts and are intuitive and easy to compute.
- Continuous features are more nuanced, needing statistical modeling using the Gaussian distribution.

CamScanner